

Why standardize?

→ Compare different distributed datasets

→ detect normality

→ detect outliers

→ Create C.I

→ Test hypothesis

→ Perform Regression Analysis

$$Z = \frac{x - \mu}{\sigma}$$

Understand the formula

We want to transform random variable from $N(\mu, \sigma^2)$ to $N(0, 1)$

Subtract mean from all obs. hence it will move the graph to origin

Pearson Correlation Coefficient: To measure linear correlation

Between 0 and 1 → +ve → When one variable changes, the other variable change in same direction

0 → No correlation → No relation ship b/w the variables

Between 0 and -1 → -ve → When one variable change, the other variable change in opposite direction

$$r_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

PDF ⇒ Probability Density Function: Probability distribution of Continuous random variable.

Probability associated with single value is always zero.

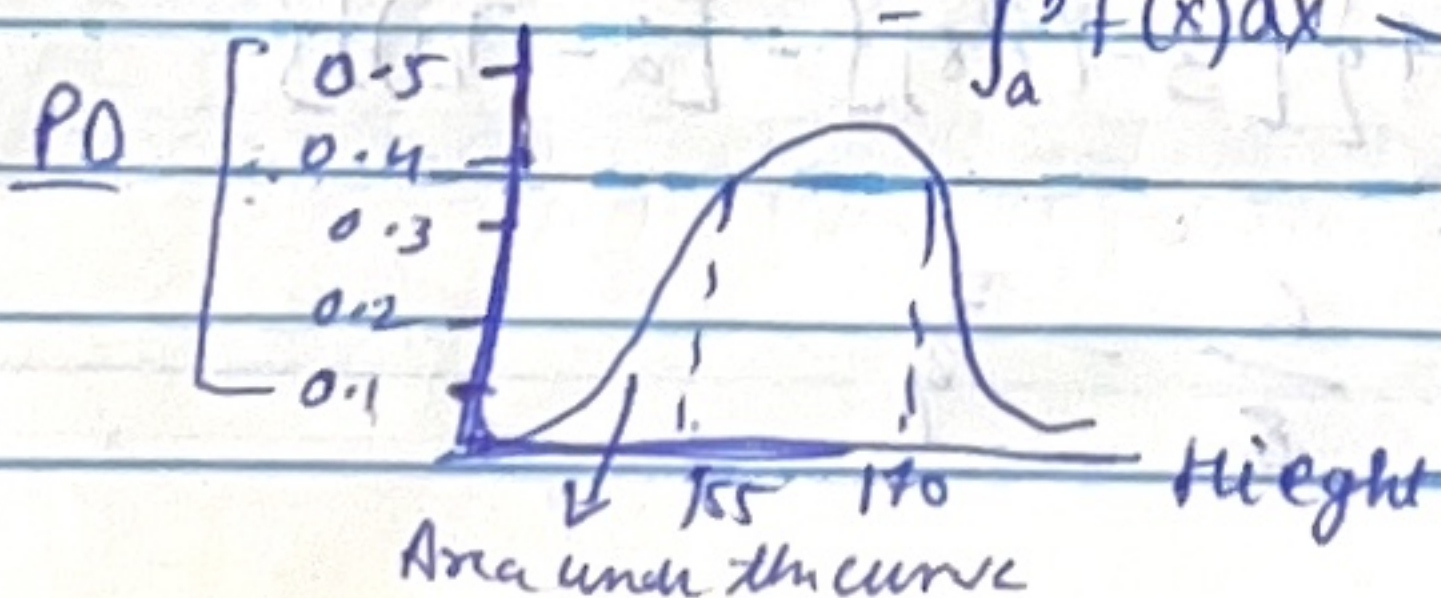
$$P_0(H \leq 155)$$

$$P(a \leq x \leq b)$$

$$= \int_a^b f(x) dx$$

$$P_0(H \geq 155 \text{ and } H \leq 170)$$

ignore end points because X is continuous



Probability / PMF → Random Variable → Discrete
 Rolling 8 Dice = $\{1, 2, 3, 4, 5, 6\}$
 Mass Function

$$\Pr(X \leq 4) = \Pr(X=1) + \Pr(X=2) + \Pr(X=3) + \Pr(X=4)$$

$$\text{Indicates Uniform distribution} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} \Rightarrow \frac{4}{6} = \frac{2}{3}$$

PDF

Let X be continuous random variable with PDF given by:-

$$f(x) = \begin{cases} x; & 0 < x < 1 \\ 2-x; & 1 < x < 2 \\ 0; & x > 2 \end{cases}$$

And $P(0.5 \leq x \leq 1.5)$

$$\Rightarrow \int_{0.5}^{1.5} f(x) dx$$

Split the integral:-

$$\int_{0.5}^1 f(x) dx + \int_1^{1.5} f(x) dx$$

Substitute the value of x :-

$$\int_{0.5}^1 x dx + \int_1^{1.5} (2-x) dx$$

$$\text{After integrating:- } \left(\frac{x^2}{2} \right)_{0.5}^1 + \left(2x - \frac{x^2}{2} \right)_1^{1.5}$$

$$\begin{aligned} \therefore & \left[\frac{(1)^2}{2} - \frac{(0.5)^2}{2} \right] + \left[2(1.5) - \frac{(1.5)^2}{2} \right] - \left[2(1) - \frac{(1)^2}{2} \right] \\ & \Rightarrow \left[\frac{1}{2} - \frac{1}{8} \right] + \left[3 - \frac{9}{8} \right] - \left[2 - \frac{1}{2} \right] \end{aligned}$$

$$\Rightarrow \frac{3+15-12}{8} = \frac{6}{8} = \frac{3}{4}$$

PMF

Let X be random Variable and $P(X=x)$

X	0	1	2	3	4	5	6	7
$P(X=x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2+K$

o) Determine value of K

o) Find probability of $P(X \leq 6)$, $P(3 < X \leq 6)$

$$0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$9K + 10K^2 = 1$$

$$10K^2 + 9K - 1 = 0$$

$$10K(K+1) - 1(K+1) = 0$$

$$(10K-1)(K+1) = 0$$

$$K = 1/10 \text{ and } K = -1 \text{ — Not possible bcz probability value ranges from 0 to 1}$$

$$P(X \leq 6) = 1 - P(X > 6)$$

$$\Rightarrow 1 - (7K^2 + K)$$

$$\Rightarrow 1 - \left[7\left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right) \right] \Rightarrow 1 - \left[\frac{7}{100} + \frac{1}{10} \right] \Rightarrow 1 - \frac{17}{100} \Rightarrow \frac{100-17}{100} = \frac{83}{100}$$

$$P(3 < X \leq 6) = P(X=4) + P(X=5) + P(X=6)$$

$$= 3K + K^2 + 2K^2$$

$$= 3\left(\frac{1}{10}\right) + \left(\frac{1}{10}\right)^2 + 2\left(\frac{1}{10}\right)^2 \Rightarrow \frac{3}{10} + \frac{1}{100} + \frac{2}{100} \Rightarrow \frac{3}{10} + \frac{3}{100} \Rightarrow \frac{30+3}{100} = \frac{33}{100}$$

CDF $\Rightarrow$ Cumulative Distribution Function
Either Continuous or Discrete

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \rightarrow \text{Continuous}$$

Random variable

$$P(X=b) = F_X(b) - \lim_{x \rightarrow b^-} F_X(x)$$

Every CDF F_X is non decreasing and right continuous

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow +\infty} F_X(x) = 1$$

For continuous random variables: $F_X(b) - F_X(a) = P(a < X \leq b) = \int_a^b f(x) dx$

X is completely discrete random variable:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i) = \sum_{x_i \leq x} P(x_i)$$

Random variable with PDF given by:

$$f(x) = \begin{cases} K(x^2 + x); & \text{if } 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$K \int_0^1 (x^2 + x) dx = 1$$

$$K \left(\frac{x^3}{3} + \frac{x^2}{2} \right) \Big|_0^1 = 1$$

$$K \left(\frac{5}{6} \right) = 1$$

$$K = \frac{6}{5}$$

If x in interval $(-\infty, x)$ then: